

A Multi-Source Retail Intelligence Platform Using Data Warehousing, Customer Segmentation, Market Basket Analysis, and Holiday-Aware Analytics

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Abstract—This paper presents a multi-source Retail Intelligence Platform built on the Dunnhumby Complete Journey grocery dataset and enriched with a U.S. Federal Holidays dataset. The platform combines Python-based data cleaning, Azure PostgreSQL data warehousing, Power BI reporting, customer segmentation, Market Basket Analysis (MBA), and a recommendation prototype. The warehouse contains a validated transaction-item fact table with 1,469,307 rows, 155,848 unique baskets, and 68,509 unique products. For MBA, the validated category-level input contains 1,078,691 basket-category rows, 155,691 baskets, and 303 product-category items. Three association rule algorithms—Apriori, FP-Growth, and ECLAT—were evaluated using an adaptive support, confidence, and lift threshold grid. The final accepted threshold was support = 0.010, confidence = 0.60, and lift = 1.50, producing 969 rules with average lift 3.1204 and maximum lift 12.1518 for all three algorithms. Since all algorithms produced identical rule-quality results at the accepted threshold, the final selection depends on engineering constraints: ECLAT is fastest at 3.47 seconds, Apriori is most memory-efficient at 3.45 MB, and FP-Growth remains a useful general-purpose alternative. Customer segmentation using RFM features compared K-Means, DBSCAN, and Hierarchical Clustering; DBSCAN achieved the highest silhouette score of 0.758 and produced three interpretable segments: VIP, Active Shoppers, and Churned customers. The results are operationalized through Power BI dashboards and an association-rule-based recommendation prototype.

Index Terms—Retail Intelligence, Data Warehouse, Azure PostgreSQL, Power BI, Market Basket Analysis, Apriori, FP-Growth, ECLAT, Customer Segmentation, DBSCAN, RFM, Recommendation System

I. INTRODUCTION

Retail organizations collect large volumes of transactional, product, household, coupon, campaign, promotion, and calendar data. However, raw operational files are usually fragmented, contain quality issues, and are not directly suitable for business intelligence or advanced analytics. A complete retail analytics solution requires a pipeline that transforms raw data into a validated warehouse, connects it to reporting tools, and then applies data mining methods to generate business recommendations.

This paper presents a Retail Intelligence Platform that integrates the Dunnhumby Complete Journey dataset [4] with a U.S. Federal Holidays dataset. The project is designed as

both a data warehouse/business intelligence system and a data mining research implementation. It supports executive sales monitoring, product performance analysis, campaign and coupon analysis, holiday-aware analysis, customer segmentation, MBA, algorithm comparison, and recommendation prototyping.

The contribution of this work is the integration of multiple components into one reproducible retail intelligence pipeline. Many academic and practical projects focus only on dashboards, only on customer segmentation, or only on association rules. In contrast, this project combines conservative data cleaning, dimensional modeling, data validation, Power BI reporting, RFM clustering, association rule mining, algorithm performance comparison, and a prototype recommendation layer.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system architecture. Section IV describes data sources and preprocessing. Section V presents the warehouse design. Section VI explains the analytical methodology. Section VII reports experimental results. Section VIII discusses business implications. Section XI concludes the paper.

II. RELATED WORK

A. Data Warehousing and Dimensional Modeling

Kimball and Ross established dimensional modeling principles based on fact and dimension tables for decision-support systems [7]. Retail analytics systems commonly use star or galaxy schemas because sales facts need to be analyzed by product, customer, store, date, campaign, coupon, and promotion dimensions. This project follows a galaxy/star-inspired model implemented in Azure PostgreSQL under the `retail_dw` schema.

B. Association Rule Mining

Agrawal et al. [1] introduced association rule mining to discover relationships between itemsets in large databases. Apriori uses candidate generation and repeated scans, which makes it transparent but potentially expensive. Han et al. [2] introduced FP-Growth, which avoids candidate generation by

compressing transactions into a frequent-pattern tree. Zaki [3] introduced ECLAT, which uses a vertical transaction-ID representation and set intersections to discover frequent itemsets. This work compares Apriori, FP-Growth, and ECLAT on the same validated retail basket data and evaluates both rule quality and engineering performance.

C. Customer Segmentation

RFM analysis summarizes customer behavior using recency, frequency, and monetary value [6]. K-Means is widely used for RFM segmentation, but it requires the number of clusters in advance and may merge outlier customers into larger clusters. DBSCAN identifies density-based clusters and can discover outlier-like groups without specifying the number of clusters manually [5]. This project compares K-Means, DBSCAN, and Hierarchical Clustering and selects DBSCAN because it gives the best silhouette score and the clearest business interpretation.

III. SYSTEM ARCHITECTURE

The platform follows a layered architecture:

- 1) **Raw Data Layer:** Complete Journey CSV files and U.S. Federal Holidays data.
- 2) **Data Engineering Layer:** Python and SQL scripts for cleaning, validation, transformation, and chunk loading.
- 3) **Staging Layer:** Cleaned files loaded into Azure PostgreSQL staging tables.
- 4) **Data Warehouse Layer:** Fact and dimension tables under `retail_dw`.
- 5) **Business Intelligence Layer:** Power BI dashboards connected to warehouse tables and validation views.
- 6) **Advanced Analytics Layer:** Python notebooks/scripts for segmentation, MBA, algorithm comparison, and recommendation outputs.
- 7) **Prototype Layer:** A shopping recommendation prototype that uses association rules to recommend products for the user's cart.

Figure 1 presents the warehouse model that was preserved as the main architectural figure in this paper. Figure 2 presents the end-to-end pipeline from raw data to dashboards, mining outputs, and the recommendation prototype.

IV. DATA SOURCES AND PREPROCESSING

A. Data Sources

The project uses the Dunhumby Complete Journey dataset, which includes transactions, products, demographics, campaigns, coupons, coupon redemptions, and promotions. A U.S. Federal Holidays dataset was added as an external enrichment source to support event and holiday impact analysis.

TABLE I
PROJECT DATA SOURCES

Source	Purpose
Transactions	Sales, quantity, baskets, households, stores, discounts, coupon usage, and transaction dates.
Products	Product hierarchy including department, category, product type, brand, manufacturer, and package size.
Demographics	Household attributes such as age, income, household size, marital status, and home ownership.
Campaigns	Campaign type, start date, end date, and campaign duration.
Coupons and Redemptions	Coupon availability and actual household redemption behavior.
Promotions	Product-store-week promotion exposure.
U.S. Holidays	Holiday names, dates, and definitions for holiday-aware analysis.

B. Cleaning Strategy

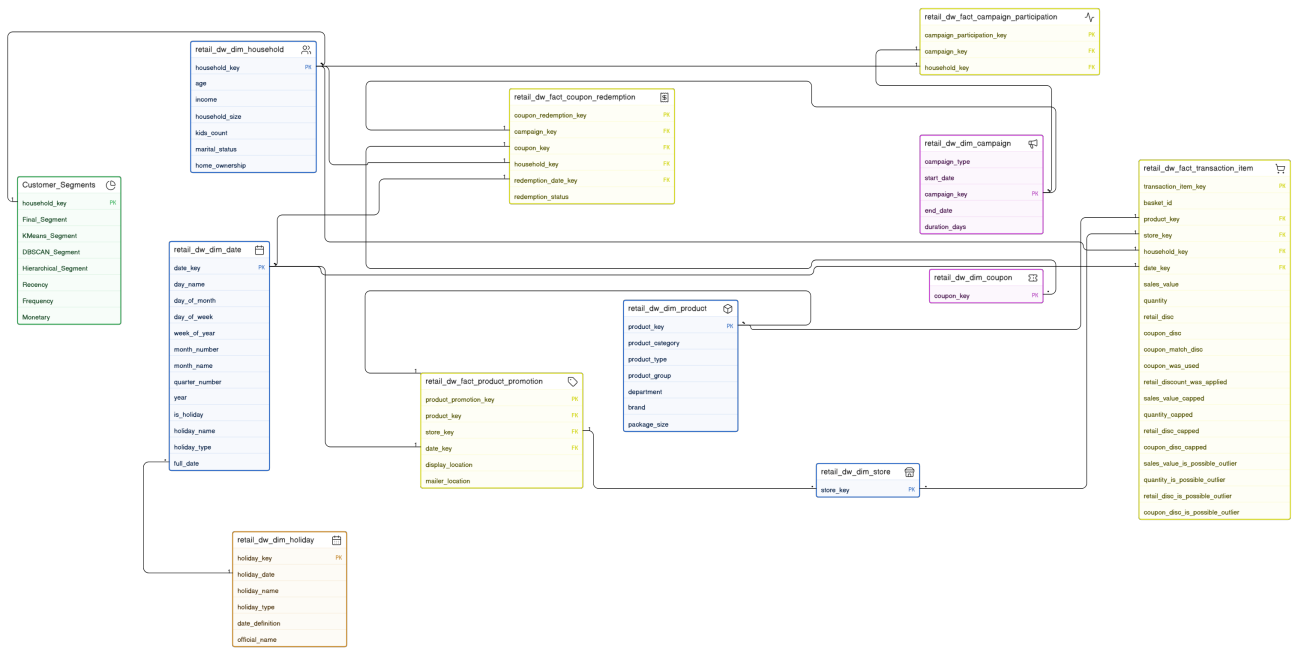
The cleaning strategy was conservative. Instead of deleting all suspicious records, the project preserved business events whenever possible and used flags, capped values, or unknown mappings. This protects sales analysis from losing important retail behavior.

TABLE II
MAIN DATA CLEANING DECISIONS

Issue	Decision	Reason
Coupon duplicates	Removed 4,872 exact duplicate rows	Prevents inflated joins and biased coupon analysis.
Invalid quantity values	Kept and flagged quantity ≤ 0 rows	May represent returns, corrections, or special retail operations.
Missing references	Kept transactions and mapped missing references to unknown dimensions where needed	Avoids damaging sales totals.
Outliers	Flagged and capped sales, quantity, retail discount, and coupon discount	Preserves traceability while improving robust reporting.
Rare coupon usage	No oversampling applied	Oversampling would distort support, confidence, and basket frequency.
Dates	Standardized transaction, campaign, and holiday dates	Enables date dimension joins and time analysis.

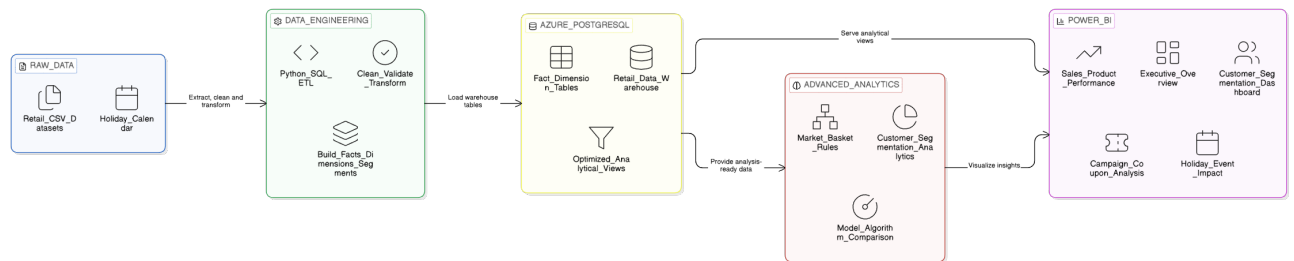
V. DATA WAREHOUSE DESIGN

The warehouse was implemented in Azure PostgreSQL under the schema `retail_dw`. It includes staging tables, dimension tables, fact tables, and validation views. The main



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Fig. 1. Retail Intelligence Platform data warehouse schema. The warehouse uses staging tables, seven dimensions, four fact tables, and validation views under the retail_dw schema. The main analytical grain is fact_transaction_item: one product purchased inside one basket by one household at one store on one transaction date.



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Fig. 2. End-to-end Retail Intelligence Platform pipeline. Raw Complete Journey and holiday datasets are cleaned and loaded into Azure PostgreSQL, modeled for Power BI reporting, and used by customer segmentation, market basket analysis, and the recommendation prototype.

fact grain is fact_transaction_item, where one row represents one product purchased inside one basket by one household at one store on a transaction date.

TABLE III
WAREHOUSE TABLE GROUPS

Group	Tables
Staging	stg_transactions, stg_products, stg_promotions, stg_holidays, stg_demographics, stg_coupons, stg_coupon_redemptions, stg_campaigns, stg_campaign_descriptions, stg_basket_items_for_mba.
Dimensions	dim_date, dim_product, dim_household, dim_store, dim_campaign, dim_coupon, dim_holiday.
Facts	fact_transaction_item, fact_coupon_redemption, fact_campaign_participation, fact_product_promotion.
Validation Views	vw_table_row_counts, vw_data_quality_summary

TABLE IV
DATA QUALITY VALIDATION RESULTS

Metric	Value
fact_transaction_item rows	1,469,307
stg_basket_items_for_mba rows	1,469,307
dim_product rows	92,349
Unique baskets for MBA	155,848
Unique products	68,509
Null basket_id	0
Null product_id	0
Duplicate basket-product pairs	0
Unmatched product references	0
Unknown product-key rows	0

VI. ANALYTICAL METHODOLOGY

A. Market Basket Analysis

MBA was applied to a category-level basket dataset. The validated input contains 1,078,691 basket-category rows, 155,691 baskets, and 303 product-category items. Product-level mining was avoided because the warehouse contains 68,509 unique products, which makes full product-level association mining computationally expensive. Category-level mining keeps the analysis practical and still supports cross-selling and recommendation scenarios.

The baskets were split deterministically into training and test sets using a hash of the basket identifier. After the 80/20 split and rare-item filtering, the training set contained 124,418 baskets, 301 categories, and 860,725 basket-item rows. Rules were mined on training baskets and evaluated as a recommender on held-out test baskets.

Three algorithms were compared: Apriori, FP-Growth, and ECLAT. The adaptive search tested support, confidence, and lift combinations from strict to loose. The accepted rule set was the strictest threshold combination that generated a rule count within the usable band of 20 to 5,000 rules. Runtime and memory were tracked for engineering comparison.

B. Recommendation Evaluation

Each test basket was split into an input half and a hidden half. Generated rules were applied to the input half, and the top five recommendations were compared against the hidden half. Recommendation quality was measured using precision@5, recall@5, hit_rate@5, and coverage. The recommendation scoring formula combined lift, confidence, and support using the following weighted score:

$$Score = 0.45 \times lift + 0.35 \times confidence + 0.20 \times support$$

C. Customer Segmentation

RFM features were computed per household: recency, frequency, and monetary value. K-Means, DBSCAN, and Hierarchical Clustering were compared. DBSCAN was selected as the final model because it achieved the highest silhouette score and identified a meaningful VIP group without requiring the number of clusters to be fixed manually.

VII. EXPERIMENTAL RESULTS

A. Basket Data Overview

The mean basket size is calculated from the category-level MBA input as 1,078,691 basket-category rows divided by 155,691 baskets, which gives approximately 6.9 categories per basket.

TABLE V
BASKET DATA OVERVIEW FOR MARKET BASKET ANALYSIS

Metric	Value
Basket-category rows before split	1,078,691
Total baskets before split	155,691
Product-category items before filtering	303
Average categories per basket	6.9
Train baskets after split/filtering	124,418
Test baskets after split/filtering	31,273
Train basket-item rows after filtering	860,725
Train categories after rare-item filtering	301

B. MBA Threshold Grid Results

Table VI reports the full tested threshold grid. All three algorithms produce identical itemset counts, rule counts, average confidence, average lift, and recommendation quality at the same thresholds. This confirms mathematical convergence: the algorithms differ in computational strategy, not in the final rule-quality result.

C. Final Accepted MBA Results

The final accepted threshold is support = 0.010, confidence = 0.60, and lift = 1.50. It is accepted because it produces 969 rules, which is inside the acceptable rule-count band of 20 to 5,000. The next looser threshold produces 5,983 rules, which is slightly above the maximum accepted rule-count band and was therefore rejected for the final production-style setup.

TABLE VII
FINAL ACCEPTED MBA RANKING

Algorithm	Rules	Avg Lift	Runtime	Memory
Apriori	969	3.1204	7.18s	3.45 MB
FP-Growth	969	3.1204	5.02s	161.18 MB
ECLAT	969	3.1204	3.47s	51.52 MB

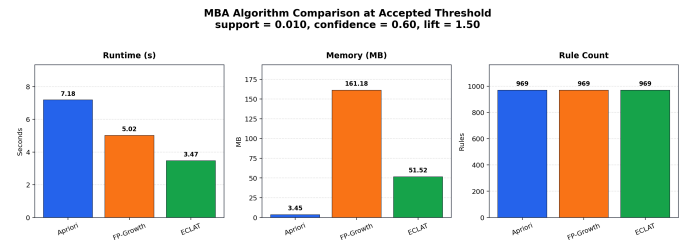


Fig. 3. Colored MBA algorithm comparison at the accepted threshold: runtime, memory usage, and rule count for Apriori, FP-Growth, and ECLAT. All algorithms produce 969 rules; ECLAT is fastest, while Apriori uses the least memory.

TABLE VI
MBA ALGORITHM RESULTS ACROSS THE THRESHOLD GRID

Algorithm	Threshold	Itemsets	Rules	Avg Conf.	Avg Lift	P@5	R@5	Hit@5	Coverage	Quality	Time	Memory
Apriori	0.010/0.60/1.50	2,381	969	0.6698	3.1204	0.166	0.048	0.246	0.027	0.381	7.18	3.45
Apriori	0.005/0.50/1.30	10,007	5,983	0.6292	3.3337	0.174	–	0.395	–	0.619	16.26	11.57
Apriori	0.003/0.40/1.20	24,452	19,727	0.5589	3.6586	0.135	–	0.431	–	0.590	30.65	26.12
Apriori	0.001/0.30/1.00	104,312	98,818	0.4535	4.2795	0.112	–	0.383	–	0.536	42.37	117.67
FP-Growth	0.010/0.60/1.50	2,381	969	0.6698	3.1204	0.166	0.048	0.246	0.027	0.381	5.02	161.18
FP-Growth	0.005/0.50/1.30	10,007	5,983	0.6292	3.3337	0.174	–	0.395	–	0.619	12.56	24.84
FP-Growth	0.003/0.40/1.20	24,452	19,727	0.5589	3.6586	0.135	–	0.431	–	0.590	19.18	30.97
FP-Growth	0.001/0.30/1.00	104,312	98,818	0.4535	4.2795	0.112	–	0.383	–	0.536	49.24	38.14
ECLAT	0.010/0.60/1.50	2,381	969	0.6698	3.1204	0.166	0.048	0.246	0.027	0.381	3.47	51.52
ECLAT	0.005/0.50/1.30	10,007	5,983	0.6292	3.3337	0.174	–	0.395	–	0.619	5.16	5.94
ECLAT	0.003/0.40/1.20	24,452	19,727	0.5589	3.6586	0.135	–	0.431	–	0.590	6.88	4.44
ECLAT	0.001/0.30/1.00	104,312	98,818	0.4535	4.2795	0.112	–	0.383	–	0.536	10.64	76.73

Threshold format = support/confidence/lift. Time is measured in seconds and memory in MB. Dashes indicate values not selected for final reporting.

D. Association Rule Quality

At the accepted threshold, all three algorithms return the same rule-quality metrics: average confidence = 0.6698, average lift = 3.1204, maximum lift = 12.1518, precision@5 = 0.166, recall@5 = 0.048, hit_rate@5 = 0.246, and coverage = 0.027.

TABLE VIII
FINAL RULE AND RECOMMENDATION QUALITY AT ACCEPTED THRESHOLD

Metric	Value
Accepted support	0.010
Accepted confidence	0.60
Accepted lift threshold	1.50
Frequent itemsets	2,381
Generated rules	969
Average confidence	0.6698
Average lift	3.1204
Maximum lift	12.1518
precision@5	0.166
recall@5	0.048
hit_rate@5	0.246
Coverage	0.027
Final quality score	0.381

E. Best Algorithm Summary

Since the rule sets are identical at the accepted threshold, the best algorithm depends on the deployment constraint.

TABLE IX
BEST ALGORITHM BY SELECTION CRITERION

Selection Criterion	Best Algorithm	Value
Final rule/recommendation quality	Tie: all three	0.381
Fastest runtime	ECLAT	3.47s
Lowest memory usage	Apriori	3.45 MB
Highest average lift	Tie: all three	3.1204
Highest maximum lift	Tie: all three	12.1518
Best hit_rate@5	Tie: all three	0.246

F. Customer Segmentation Results

Table X compares the three segmentation algorithms. DBSCAN achieved the strongest silhouette score and produced the most useful business interpretation.

TABLE X
CUSTOMER SEGMENTATION ALGORITHM COMPARISON

Algorithm	Silhouette	Interpretation
K-Means	0.656	Clean separation, mainly Active vs. Churned.
DBSCAN	0.758	Final selected model; identifies VIP outliers.
Hierarchical	–	Baseline comparison supporting segment structure.

TABLE XI
FINAL DBSCAN CUSTOMER SEGMENTS

Segment	Size	Recency	Frequency	Monetary
Active Shoppers	2,431 (97%)	22 days	103 baskets/year	\$3,080
VIP	55 (2.2%)	83 days	476 baskets/year	\$10,304
Churned	14 (0.6%)	544 days	4 baskets/year	\$163

G. Power BI Dashboard Results

The Power BI dashboard includes pages for executive overview, sales and product performance, customer segmenta-

tion, campaign/coupon/promotion analysis, holiday and event impact, and data warehouse validation.

TABLE XII
MAIN POWER BI DASHBOARD PAGES

Page	Purpose
Executive Overview	High-level sales, baskets, average basket value, active households, and coupon usage.
Sales and Product Performance	Department, category, product, and sales trend analysis.
Customer Segmentation	VIP, Active Shoppers, and Churned customer profiles.
Campaign and Coupon Analysis	Campaign participation, coupon redemptions, and discount behavior.
Holiday and Event Impact	Holiday vs. non-holiday sales and coupon behavior.
Data Quality Validation	Row counts, validation checks, and warehouse trust indicators.

VIII. KEY INSIGHTS AND BUSINESS IMPLICATIONS

A. MBA Algorithm Selection

The main MBA finding is that Apriori, FP-Growth, and ECLAT produce identical rule-quality results at the same threshold. Therefore, algorithm selection should be based on runtime and memory constraints. ECLAT is recommended when low runtime is the priority because it completed the accepted run in 3.47 seconds. Apriori is recommended when memory usage is the main constraint because it used only 3.45 MB. FP-Growth remains a valid general-purpose option, but in the accepted run it consumed the highest memory.

B. Threshold Selection

The accepted threshold of support = 0.010, confidence = 0.60, and lift = 1.50 gives an interpretable rule set of 969 rules. Looser thresholds such as support = 0.005 and confidence = 0.50 achieve better recommendation hit_rate@5, but they generate 5,983 rules, which is above the selected operational rule-count band. This creates a clear trade-off between recommendation coverage and rule-set manageability.

C. Recommendation Interpretation

At the accepted threshold, hit_rate@5 = 0.246 means that approximately one out of four test baskets received at least one top-five recommendation that appeared in the hidden part of the basket. precision@5 = 0.166 means that around 16.6% of recommendation slots were correct. Coverage = 0.027 means that the strict threshold focuses recommendations on a small set of reliable categories rather than spreading recommendations across the whole catalog.

D. Customer Segmentation Value

DBSCAN produced three business-friendly groups. Active Shoppers are the core customer base and should receive personalized category promotions. VIP customers represent a small but high-value group with high frequency and monetary value, so they should receive loyalty and retention treatment.

Churned customers have very high recency and low frequency, so they are suitable for win-back campaigns.

E. Campaign, Coupon, Promotion, and Holiday Analysis

The platform connects sales behavior with campaigns, coupons, promotions, and holidays. This allows business users to analyze which campaigns attract households, how often coupons are redeemed, which categories receive discounts, and whether holidays change sales behavior. The holiday dimension was merged with the date model to avoid incorrect cross-join behavior in Power BI.

IX. LIMITATIONS

The project has several limitations. Product-level MBA was not used in the final experiment because 68,509 unique products make full item-level mining expensive. The final MBA uses category-level rules, which are more scalable but less granular. Some household demographics are available only for a subset of households. Customer segments need business validation before operational use. The recommendation prototype is a demonstration layer and is not yet a production real-time recommender.

X. FUTURE WORK

Future work includes testing product-level MBA using distributed or approximate mining, adding collaborative filtering for personalized recommendations, automating ETL and Power BI refresh, deploying the recommendation prototype, adding external event datasets such as weather and regional events, improving campaign ROI analysis, and adding churn prediction models.

XI. CONCLUSION

This paper presented a Retail Intelligence Platform that integrates data engineering, Azure PostgreSQL warehousing, Power BI dashboards, customer segmentation, MBA, algorithm comparison, and recommendation prototyping. The project produced a validated warehouse with 1,469,307 transaction-item rows and a validated MBA input suitable for association rule mining. The final accepted MBA threshold was support = 0.010, confidence = 0.60, and lift = 1.50, producing 969 rules with average lift 3.1204 and maximum lift 12.1518. Apriori, FP-Growth, and ECLAT tied on rule quality, while ECLAT was fastest and Apriori used the least memory. DBSCAN was selected for customer segmentation because it achieved the highest silhouette score and produced interpretable VIP, Active Shopper, and Churned segments. Overall, the platform provides a complete foundation for retail decision support, targeted promotions, cross-selling, holiday-aware planning, and future recommendation systems.

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